

Business Question and Visualization Report

Global AI Adoption in Education

Field	Detail
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Project Name	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Team ID	Individual
Tools	Tableau Desktop / Tableau Public

Visualization development transforms raw data into graphical formats that facilitate understanding, analysis, and decision-making. For this project, Tableau was used to convert **1,360 monthly records (2015–2026) across 10 countries, 6 regions, and 16 attributes** into 8 focused worksheets, 1 consolidated dashboard, and a 5-page story. The visualizations enable stakeholders — educators, policymakers, and researchers — to gain insight into global AI adoption trends, equity gaps, and the relationship between policy and actual classroom usage.

Business Questions and Visualisation

1. What are the overall key performance metrics of global AI adoption in education?

- **Visualization:** Text Table (KPI Summary) showing Avg. Student AI Usage: 30.99% | Avg. Internet Penetration: 74.70% | Avg. Education Index: 0.776 | Curriculum Integration Rate: 50.51% (687 of 1,360 records).
- **Purpose:** To provide an overall snapshot of the global dataset before users explore detailed visualizations.

2. How has AI adoption grown across students, teachers, and schools from 2015 to 2026?

- **Visualization:** Multi-line time series showing student_ai_usage_pct, teacher_ai_usage_pct, and schools_ai_adoption_pct over time.
- **Observation:** All three adoption layers rise together in near lock-step — from roughly 4.98% (student) in 2015 to 60.75% by 2026, a roughly twelvefold increase — with teacher (59.00%) and school-level (59.25%) adoption converging closely behind by the final year.
- **Purpose:** To visualize the trajectory of AI adoption and confirm whether growth is consistent across the three stakeholder layers.

3. Are some countries meaningfully outpacing others in AI adoption?

- **Visualization:** Horizontal bar chart ranking countries by average student_ai_usage_pct.

- **Observation:** Adoption is remarkably uniform globally — bar lengths range only from 30.60% (Brazil) to 31.61% (United States), a spread of about one percentage point. No country is a clear runaway leader or laggard.
- **Purpose:** To test the assumption that certain countries dominate AI-in-education adoption — and disprove it with the data.

4. Does internet penetration or education quality predict how fast a country adopts AI?

- **Visualization:** Scatter plot with trend line comparing `internet_penetration_pct` and `education_index` against `student_ai_usage_pct`.
- **Observation:** Internet penetration and education index correlate strongly with each other ($r \approx 0.98$, as expected), but neither correlates meaningfully with student, teacher, or school AI usage ($r \approx 0.00$ in all cases) — adoption is being driven primarily by time/global trend, not digital infrastructure or education-system quality.
- **Purpose:** To directly challenge the assumption that better-connected or higher-scoring education systems adopt AI faster.

5. How large is the urban-rural AI usage gap, and is it closing over time?

- **Visualization:** Dumbbell / diverging bar chart showing Urban vs. Rural AI usage % by country, alongside a year-over-year gap trend line.
- **Observation:** The gap widened sharply from 7.68 points in 2015 to roughly 12–13 points by 2018, and has stayed essentially flat since (12.98 points in 2026). By country, Pakistan (12.24 pts) and Canada (12.22 pts) show the widest gaps; India (11.57 pts) the narrowest.
- **Purpose:** To test whether overall adoption growth is closing the urban-rural equity gap — it is not.

6. Does a formal government AI policy translate into curriculum integration?

- **Visualization:** Stacked bar chart comparing `government_ai_policy` status (Active / Draft / None) against the % of records with `ai_in_curriculum` = Yes.
- **Observation:** Curriculum integration sits at 52.3% under an Active policy, 51.9% with no formal policy at all, and only 47.2% under a Draft policy — a strikingly flat relationship, showing a formally “Active” policy is a weak predictor of real classroom integration.
- **Purpose:** To evaluate whether policy is driving or merely coinciding with classroom adoption.

7. Which AI tool dominates in each region?

- **Visualization:** Stacked bar chart / treemap showing `top_ai_tool` share by region.
- **Observation:** Asia and Africa show the broadest four-way split among ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo. The most recent months of 2026 (a smaller four-month sample) show ChatGPT pulling ahead — a trend worth monitoring rather than treating as fully confirmed given the limited sample size.
- **Purpose:** To understand the competitive landscape of AI tools across different educational markets.

8. Is there a gender gap in AI usage, and is it closing?

- **Visualization:** Line chart showing gender_gap_ai_usage_pct by year, 2015–2026.
- **Observation:** The gender gap has held steady between 5.05 and 5.91 points across the entire period, with no clear narrowing trend — mirroring the persistence seen in the urban-rural gap.
- **Purpose:** To monitor equity in AI access between genders and check whether growth in overall adoption is translating into a more equitable outcome.

Summary Takeaway

Across all eight business questions, one theme recurs: **AI adoption in education has grown enormously (roughly twelvefold in 12 years) but almost entirely uniformly** — across countries, regardless of infrastructure, and regardless of formal policy status. The equity gaps present at the start of the dataset (urban-rural, gender) have not closed as adoption has scaled; they have simply persisted alongside it. This is the central, decision-relevant insight for policymakers and edtech stakeholders reviewing this analysis.